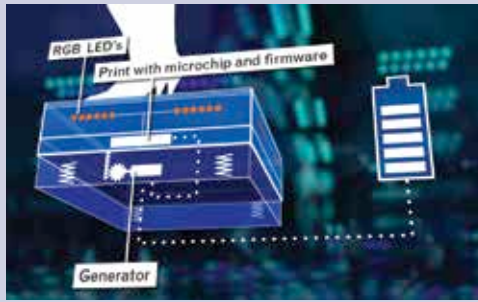


LET'S GET PHYSICAL

Piezoelectricity was discovered in the late 1800s by French physicists and brothers Pierre and Jacques Curie. Yes, Pierre, husband to Marie — the Nobel Prize winning pair for radioactivity. Certain materials, when put under physical stress, produce an electrical charge. This is best seen in electric cigarette lighters: hit a button, which hammers a piezoelectric crystal and produces a spark. Stretching and bending such materials can also result in electrical activity, which can then be used as a generator. One such application foresees sheets of piezoelectric material woven into the soles of a shoe. As you walk, the material bends and stretches, and the electricity produced is stored in small batteries. There have been a few practical uses to this tech: a dance club in the Netherlands, appropriately dubbed WATT, uses piezoelectric modules to harness the dancers' steps and power LED lights by the dance floor. Each person can generate about five to twenty Watt. Each module is 75x75x20 cm and can produce up to 35 watt of sustained output.



The reality was bleaker. Or it could have been, if given a chance to manifest. While millions were sold on the pill, millions still were skeptical, or worse. The detractors pointed to several unfortunate incidents in the past; including everyone's favourite that involved — a school bus full of children and a malfunction with the heads-up display of the bus driver that set the opacity of an incoming message too high. The resultant accident put a severe deceleration on the development of HUDs; it was only with some extremely stringent policies (and massive global lobbying) introduced by Odin Werks that solutions such as OdinVision were allowed back into the consumer market.

The marketing pitch, never realised into an actual product: starting with the US FDA, the miracle pills were banned across countries. The Package that I was expecting would contain some sensitive information on the nanopills. More

precisely, it contained the location of the location, which would give me an indication to the where and whom of the pills. I focused on the package and the house's cameras zoomed in on the label, the OdinVision showing me the feed as an overlay on the left side of my vision. The packaging didn't give me any clues except to show a generic destination and source, along with the route that the package took to get here. It could be anything. I would have to get up and get my hands dirty. Sigh. I groaned, stepped onto the floor and dragged my feet towards the door. Each step that I took pressed against a piezoelectric pad below the flooring and added a little bit of power to the home's battery storage. Walking from my room to the hall, in order to open the door, would generate enough power to do absolutely nothing. But it all added up: the pressure pads in the floor, the solar panels in the window, the wind vanes

on the terrace, the house also borrowed power generated by commuters walking below — pads, like the ones used in my floor would funnel wattage from passing feet to a nearby power routing lamp-post, meanwhile multiple vents along the building would suck in the collective heat of the billion Mumbaikars and transform it to energy using good old-fashioned water and steam. Energy was never more symbiotic and communal.

As I reached for the door, I noticed that the refrigerator in the kitchen was continuing its attention-grabbing “dude, you are all out of food” notification. It had been doing this all week. Normally, the fridge had enough sense to order my usual meats and vegetables — it pretty much set my diet, after all. This week, however, was different. My credit flow was ebbing; not nearly enough for the refrigerator to directly place an order to the hypermarket. That would need my intervention, which wasn't forthcoming, and hence the notification. I could add some funds to the house's wallet but that would immediately be siphoned off to the rent first and because I hadn't paid for the past two months, it would not leave anything for food. No, I would have to do this the hard way as well, and go to the hypermarket myself to grab some bits and pieces to eat.

I opened the door to the usual sights and smells of the city.

One sight in particular was especially arresting: my neighbor coming back from her morning jog. I trained my OdinVision on her and took a few snaps for memorabilia. If she were paying attention, she would have noticed my soft lenses strobe a visible red — this was a deliberate feature of OdinVision — one necessary to communicate to a party that he/she is being watched and recorded. It was socially important to broadcast this form of communication, much like the shutter sound effects in camera phones of yore... “You are on film”, it said. A few snaps for my scrapbook, harmless. But the body can't be fooled and by extension, neither can OdinVision. The slight uptick in endorphins that the sight of my neighbor triggered was noted by OdinVision and sure enough, I promptly got a surge of advertisements: interest you in flowers perhaps?, a few local ‘romantic spots’ that you two should surely check out, an ad for chocolates (always works), and of